2.3.2 Stress and Strain

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Process

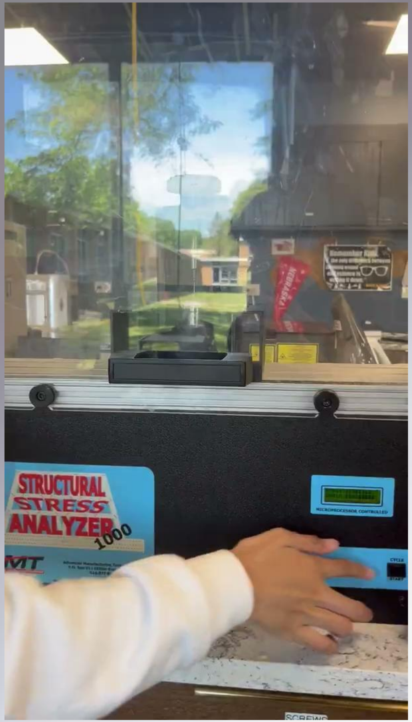
We had to find what type of material the three dog bones were by having each of them undergo a tensile test. After getting the data from the tensile test we figured out what type of material each dog bone were by using math.

Math

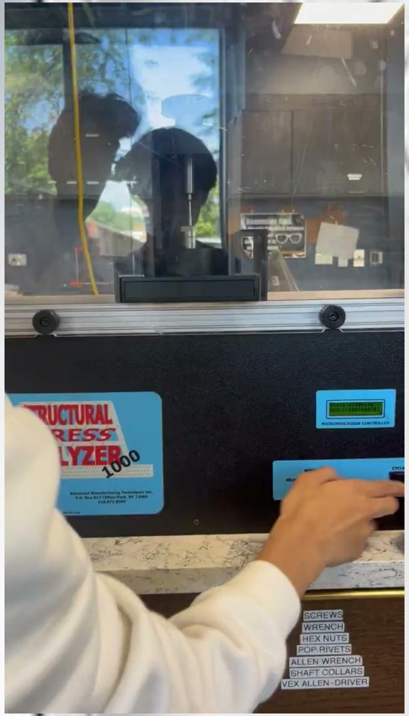
A	B	C	D
	Brassish	Steelish	Tungstenish
Original Length (Lo)	2	2	2
Change in Lenth (ΔL)	0.07	0.09	0.06
Normal Strain (ε)	0.035	0.045	0.03
Axial force (P)	54510.56801	82362.68305	29244.72079
Radius (r)	0.04	0.04	0.04
Diameter (d)	0.08	0.08	0.08
Cross-section area (A)	0.00502655	0.00502655	0.00502655
Stress (σ)	54510.56801	82362.68305	29244.72079
Shear stress (τ)	10844532.94	16385535.17	5818052.342
Modulus of elasticity (E)	1557444.8	1830281.846	974824.0264
Force (F)	274	414	147

Videos

Dogbone #1 F = 274 lbs Displ = 0.035 in



Dogbone #2 F = 414 lbs Displ = 0.023 in



Dogbone #3 F = 147 lbs Displ = 0.015 in

R/W/L

Right	Wrong	Learned
• We did the actual experiment very quickly, leaving us with time to finish everything else. • We found the metals of the dogbones.	• We slipped up when using the breaker a couple of times, causing us to have to redo one of the dogbones	• We learned new math and gained experience with different ways to use the breaker.